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Mixture of experts framework for continuous contactless dynamic blood pressure estimation from radar signal

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ABSTRACT

Continuous and unobtrusive dynamic blood pressure monitoring is pivotal for the diagnosis and prevention of cardiovascular diseases. However, the inherent instability in contactless vital sign signal acquisition and medical resource constraints presents formidable challenges to the robustness and computational load of models. In this study, the Mixture of Experts Feature Fusion Neural Network (MoEF-Net) is introduced, an approach in which the flexibility and modularity of the metaformer architecture are applied for the extraction of physiological signal features and the estimation of dynamic blood pressure. Distinct from existing methods, MoEF-Net effectively embeds variable time-domain waveform information of physiological signals by integrating expert models and gating mechanisms. During the feature extraction stage, multi-scale convolutional modules and pixel-level gating mechanisms enable dynamic weight adjustment of the embedded feature maps, thereby facilitating robust feature representation. The proposed MoEF-Net framework achieves robust blood pressure estimation, with the original model attaining mean absolute errors (MAEs) of 4.64 mmHg for systolic and 3.58 mmHg for diastolic pressure. To enhance practical deployment, the lightweight variant MoEF-Net$_L$ was developed. It maintains highly competitive accuracy with MAEs of 4.74 mmHg and 3.71 mmHg, while its primary advantage lies in a substantial reduction in computational cost and inference time, making it exceptionally suited for resource-constrained clinical environments. The overall performance achieved Grade A under the British Hypertension Society (BHS) protocol.

1. Introduction

The blood pressure monitoring demand in modern medicine has increased, as it serves as a crucial indicator of cardiovascular function, and effective monitoring aids in the prevention and diagnosis of cardiovascular diseases [1–3]. Cuff-based blood pressure monitors present limitations, including discontinuous measurements, discomfort during prolonged use, and unsuitability for burn patients [4]. Previous studies have demonstrated the transformative potential of continuous noninvasive hemodynamic monitoring technologies in redefining blood pressure assessment methods [5]. The dynamic blood pressure can be estimated through pulse wave monitoring by extracting and analyzing pulse wave features, which are rich in hemodynamic information [6, 7]. Cuffless blood pressure estimation algorithms typically establish a dynamic relationship with real-time blood pressure by calculating pulse wave transit time (PTT), pulse wave velocity (PWV), or pulse arrival time (PAT) [8–10]. Pulse waves resulting from periodic cardiac pulsations have become essential biomarkers for the early screening of cardiovascular diseases [11–13]. By modeling the relationship between pulse wave characteristics and corresponding dynamic blood pressure, blood pressure monitoring is transformed into pulse wave monitoring. However, monitoring pulse waves is limited to contact-based photoplethysmography (PPG) methods. Although this approach provides relatively stable signals, limitations remain for the long-term monitoring of subjects.

Recent studies have demonstrated that radar technology can monitor human movements, including subtle surface motions such as respiration and heartbeat, through the analysis of time and frequency information in a non-contact manner [14–17]. These surface micro-movement signals reflect the activity of the lungs and heart, respectively. Moreover, a correlation has been identified between cardiopulmonary functions, such as heart rate variability, respiratory rate, and blood pressure [18–20]. The application of radar technology to monitor micro-movements of the chest skin surface has been demonstrated,

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facilitating the extraction of pertinent features and the development of a model that correlates this data with real-time blood pressure measurements [21–24]. Millimeter-wave radar, with its high precision and capability for direct detection of cardiac micro-movement characteristics, enables the measurement of Chest Pulse Micro-Vibration (CPMV) induced by cardiac motion, facilitating contactless monitoring of hemodynamic parameters [25]. Data utilized in this study were derived from multiscale feature information of the same radar echo signals, which have previously been demonstrated to reliably extract hemodynamic information through time-domain waveform analysis [26].

Existing radar-based contactless heartbeat micro-movement measurement methods are generally categorized into two approaches. The first approach involves measuring micro-movements of the radial artery at the wrist, which offers the advantage of being free from respiratory component interference [27,28]. However, physiological activity at the wrist is challenging to control or quantify, and precise alignment with the radial artery pulse position is required. The second approach involves measuring micro-pulse movements at the chest, typically including respiratory and heartbeat signals. This method necessitates additional filtering to address the presence of respiratory components [29,30].

During data preprocessing, physiological signals were derived from computational processes applied to demodulated radar echoes. The chest micro-movement signals, collected via radar, are subjected to filtering with frequency-specific parameters to isolate the CPMV and Radar-collected heartbeat (RHB) signals. CPMV and PPG signals share similar waveform characteristics induced by periodic cardiac pulsation in hemodynamic assessments. The second derivative of the PPG signal, known as the acceleration pulse wave, has been validated as an indicator of arterial stiffness [31,32]. To optimize the model inputs, the same processing was applied to the CPMV signals, resulting in the second derivative of CPMV (SCPMV). By treating the combined pathways of electrocardiogram (ECG) and PPG as graph information, real-time feature extraction and blood pressure estimation have demonstrated robust performance in cuffless, contact-based blood pressure monitoring [33]. Due to the limitations of radar sensors, the non-contact acquisition method precludes the collection of ECG and PPG signals. Inspired by this effective data input strategy, CPMV, RHB, and SCPMV are treated as three-channel feature-map inputs, enabling the model to focus on the detailed characteristics of the temporal waveforms.

The Mixture of Experts Feature Fusion Neural Network (MoEF-Net) was developed for feature extraction and blood pressure estimation, utilizing a metaformer-based architecture as the backbone [34–36]. The architecture performs flexible fusion of token features within a separable framework that improves modularity and clarifies the roles of mixing and projection. On this basis we present an end to end radar pipeline with temporally coregistered CPMV, SCPMV and RHB inputs. The backbone integrates mixture of experts routing and pixel level gating to learn morphology aware representations, and we provide a lightweight variant for embedded deployment. Under the British Hypertension Society protocol the method attains Grade A. In the Mixture of Experts (MoE) stem module, primary feature extraction of physiological signals is adaptively adjusted by integrating a progressive feature extraction structure with an expert system. The expert system captures diverse biomedical feature patterns by employing the gating mechanism to weight and combine outputs from various experts, allowing dynamic adjustment of feature maps [37]. This process corresponds to embedding the original input. The routing mechanism is driven by a window level context vector derived from each input segment, enabling experts to specialize in distinct pulse-wave regimes, including the rapid systolic upstroke, the diastolic runoff, and reflection-related microstructures. Feature extraction in MoEF-Net is anchored by the Multi-scale Adaptive (MsA) gated convolutional module and the MoE downsample block. Within the MsA module, feature weights are modulated through a gating pathway that adjusts dynamically across temporal scales, thereby enhancing representational capacity and training efficiency. The MoE downsample integrates channel refinement with spatial compression to produce higher-order abstractions. Through this hierarchical design, the network attends to waveform specific characteristics of the signals and captures hemodynamic patterns that underpin blood-pressure estimation from complex physiological dynamics. The main contributions of this paper are summarized as follows:

- CPMV, SCPMV, and RHB signals, verified to carry hemodynamic information, are used as a three-channel feature-map input after temporal co-registration. This formulation preserves waveform morphology, emphasizes per-channel temporal characteristics, and enables complementary cues across channels to be exploited for fine-grained temporal detail.
- A modular metaformer architecture is employed, with initial feature encoding guided by an MoE stem. Through trainable expert and gating units, this module performs adaptive feature extraction and modular token fusion, promoting pathway specialization to distinct spectral and temporal patterns and clarifying the methodological contribution beyond dataset choice.
- During feature extraction, an MsA Gated CNN together with an MoE downsampling module is applied to accommodate complex and variable physiological signals. Multi-scale convolutions and expert mechanisms provide effective representation under constrained computation, while a pixel-level gating mechanism dynamically reweights features to sharpen salient structures and improve optimization stability; ablation analyses substantiate the advantage of these components.
- A highly efficient lightweight version of MoEF-Net (MoEF-Net$_L$) is developed and recommended as the primary model for practical deployment. By strategically rearranging the feature-extraction stages, it reduces the computational cost and inference time by over 84% and 57% respectively. This exceptional trade-off establishes MoEF-Net$_L$ as the superior choice for resource-constrained clinical environments.

2. Model architecture

2.1. Overview of MoEF-Net

An end-to-end approach was employed in designing the overall architecture of MoEF-Net, incorporating the modular design principles of metaformer. By integrating the flexible characteristics of this structure and accounting for the inherent instability of vital sign input signals, multi-channel feature representation performance is enhanced under real-time input conditions. In the initial MoE stem module, three homologous multimodal vital sign signals are processed as three-channel feature information. After the MoE stem module processes the input, the resulting feature maps are fed into the MsA gated CNN module for further refinement. Subsequently, the MoE downsample module is applied to progressively reduce the spatial dimensions while enhancing feature abstraction, ensuring effective capture of subtle signal variations throughout the pipeline. Furthermore, a multi-scale feature fusion strategy is incorporated into the proposed structure to enhance model robustness when processing complex scenarios. This modular design reduces computational costs while preserving feature representation capabilities. The proposed model effectively captures subjects' hemodynamic information for real-time dynamic blood pressure estimation. It relies solely on the multimodal features of current physiological signals without dependence on long-term temporal information.

Module choices are justified by pulse-signal characteristics and by how the learning objective shapes the representation. The MoE stem builds a window-level context and learns routing weights by backpropagation so that expert paths receive higher weight on inputs that reduce the training loss; this encourages experts to specialize to distinct temporal and spectral regimes such as rapid systolic upstroke, slower diastolic runoff, and reflection related microstructure and it reduces interference among heterogeneous patterns. The MsA gated CNN applies parallel one dimensional convolutions at multiple kernel sizes to represent features over short to intermediate temporal scales. The position dependent gating then performs elementwise modulation of the concatenated features, producing per time and per channel weights that are learned to minimize the prediction error; the squeeze excitation block computes channel coefficients from a global summary and rescales the channels accordingly, which lets the network attenuate components that are not predictive under the loss and amplify those that are. The MoE downsample stage uses stride to reduce temporal resolution and expand channels, which increases the receptive field and supports a coarse to fine separation of slower envelopes and faster cardiac micro motions that are refined again by later MsA blocks. The output head maps the resulting morphology aware representation to systolic and diastolic blood pressure through fully connected layers trained end to end with the same objective. MoEF-Net is organized into four feature extraction stages. The first stage employs a MoE stem module, which leverages multi-scale expert routing to adaptively capture cross-channel representations of pulse-wave dynamics across different temporal scales. The resulting feature maps are subsequently processed by the MsA gated CNN module. As depicted in Fig. 1, the numbers of MsA gated CNN modules at different stages, denoted as $\mathbf{M}_{n1}$ to $\mathbf{M}_{n4}$, are set to (1, 3, 9, 1) in MoEF-Net. The output channel numbers for each stage are (24, 48, 96, 144), respectively. The configuration of parameters was selected based on extensive hyperparameter tuning to achieve an optimal balance between feature extraction capability and computational efficiency, with a focus on enhancing sensitivity to temporal variations in physiological signals. A structured hyperparameter search finalized the operating point. Candidates used identical preprocessing and optimization, and selection optimized joint MAE for SBP and DBP under runtime and memory constraints. The search varied the expert count in dynamic convolutions, the multi scale kernel set, and the distribution of stage depth and channel width along the pyramid. Accuracy increased with expert count until saturation, after which gains were marginal while cost grew linearly. For the MsA Gated CNN module, ablation confirms the necessity of the multi scale design. Removing the 7 × 7 branch slightly increases MAE because it restricts the receptive field needed to model reflected waves and slow morphology. Expanding the receptive field to 9 × 9 also reduces performance and increases computational cost. Capacity of the model was placed where temporal resolution remains informative and early noise is attenuated, so depth was concentrated in the middle stages and kept shallow at the entrance and exit to limit latency. Channel width increased as temporal resolution decreased to preserve representational power and to align with common hardware vectorization. A lightweight variant retained the mid stage capacity pattern, reduced early width, and redistributed depth toward the middle, which preserved most accuracy while reducing latency and memory.

The feature extraction stages are based on the feature pyramid model, resulting in reduced spatial dimensions of feature maps as the number of channels increases [38]. Specific parameters are provided at the lower left corner of each stage; for example, in stage 1 (1, 24), one MsA gated CNN module with 24 output feature channels is included. Details of the MoEF-Net model parameters in first extraction stage and output module are provided in Table 1; examples of MoE convolution and fully connected (FC) parameters are illustrated in the bottom right corner. The MoE stem performs learned routing that encourages expert specialization to distinct spectral and temporal regimes, reducing interference between heterogeneous radar patterns and improving sample efficiency. The MsA-gated CNN combines multi-scale convolutions with pixel-level weighting so that salient ridge-like structures are enhanced while background variations are suppressed, which stabilizes optimization under contactless signal instability. Together with the MoE downsampling module, the backbone maintains representation fidelity under bounded computation. A lightweight variant rearranges the depth of intermediate stages and compresses channels while preserving the high-dimensional bottleneck at the final stage, which yields lower parameter count, lower giga floating-point operations, and millisecond-level latency with minimal accuracy loss. This preserves the end-to-end pipeline for embedded and bedside scenarios.

2.2. Mixture of experts stem module

The MoE stem module is the front-end processor for physiological signals, receiving multi-dimensional feature maps concatenated along the channel dimension. This structure comprises two MoE convolution modules, Layer Normalization, and an activation function. The MoE convolution module defines multiple convolution operations as expert models within an adaptive computational process. Expert outputs are dynamically selected and combined through a gating mechanism, where the gating model is trained using a cross-entropy-based loss to ensure effective differentiation between experts. This enables each expert to specialize in a distinct feature domain, optimizing overall feature extraction performance for varying input conditions. The parameters of the model include the input channel number C_{in}, output channel number C_{out}, output sequence length H' and the number of experts N_e. As a result, there are N_e distinct convolution operations, each module corresponding to a standard convolution computation performed by an expert module.

The MoE stem module features multiple adaptive convolutional pathways. Each expert convolution layer in this module is trained to specialize in a specific aspect of input data variability, such as high-frequency noise or subtle temporal patterns. The gating mechanism adaptively selects and combines these experts based on the characteristics of the input, achieving more efficient feature extraction under varying physiological signal conditions. The computation process is as follows:

The input three-channel signal features are defined as $\mathbf{x} \in \mathbb{R}^{C_{\text{in}} \times H}$, where $C_{\text{in}} = 3$ represents the number of input signal channels including the CPMV, SCPMV and RHB. The H corresponds to the length of each input signal, which is uniformly resampled to 512 for consistency. By applying average pooling along the temporal dimension, the original input $\mathbf{x}$ is reduced to $\mathbf{c} \in \mathbb{R}^{C_{\text{in}}}$, representing the global information of each input sample. The N_e is defined as the number of experts in the MoE module. The gating function is then used to predict the expert weights $\mathbf{g} \in \mathbb{R}^{N_e}$ based on the input data vector $\mathbf{c}$:

$$\mathbf{g} = \operatorname{softmax}\left(W_g \mathbf{c} + b_g\right), \tag{1}$$

where $W_g \in \mathbb{R}^{N_e \times C_{\text{in}}}$ and $b_g \in \mathbb{R}^{N_e}$ are learnable parameters of the gating function. Through the softmax operation, the weights $\mathbf{g}$ are normalized, ensuring that the sum of the expert weights for each sample equals 1. The C_{out} and H' correspond to the output channel number and spatial dimension of each expert module. The output of each expert convolution module $\mathbf{y}_i \in \mathbb{R}^{C_{out} \times H'}$ is weighted by the corresponding coefficient g_i, resulting in the final output $\mathbf{y}$:

$$\mathbf{y} = \sum_{i=1}^{N_e} g_i \cdot \mathbf{y}_i. \tag{2}$$

Here, g_i represents the weight computed through a gating mechanism, which is expanded to match the output dimensions, and applied to the output at each spatial location. The final output $\mathbf{y} \in \mathbb{R}^{C_{out} \times H'}$ is the weighted sum of the outputs from expert modules.

2.3. Multi scale adaptive gated convolution module

The MsA gated CNN module comprises multiple computational processes, including multi-scale convolution, feature fusion, normalization, and attention mechanisms. The input feature map corresponds to the output of the MoE stem or MoE downsample modules, denoted as $\mathbf{y} \in \mathbb{R}^{C_{out} \times H'}$.

Initially, the input data undergoes Layer Normalization (LN) to normalize the feature map and stabilize the training process. The

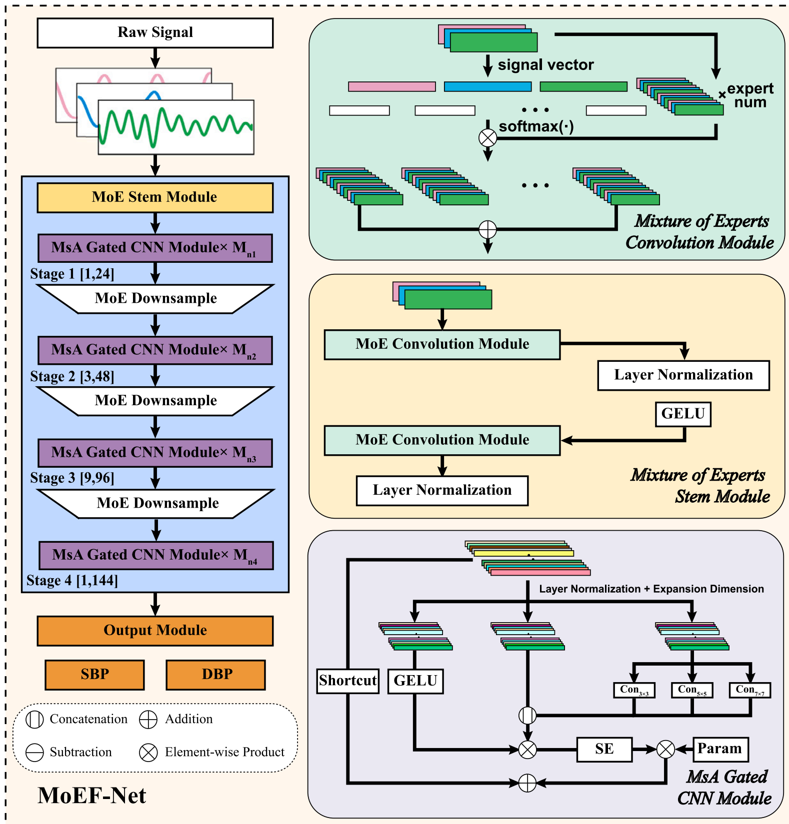

Fig. 1. The overall architecture of MoEF-Net consists of four feature extraction stages, with the right panel illustrating the specific computational processes of each module.

Table 1
Architectural parameters details of the MoEF-Net.

Module	Module type	Input size	Output size
MoE Stem Module	MoE Convolution (6,12,2)	(3, 512)	(12, 256)
	LN	(256, 12)	(256, 12)
	GELU	(12, 256)	(12, 256)
	MoE Convolution (3,24,2)	(12, 256)	(24, 128)
	LN	(128, 24)	(128, 24)
MsA gated CNN Module	FC (24,128)	(128, 24)	(128, 128)
	Split Feature Channel	(128, 128)	(64, 128), (40, 128), (24, 128)
	$\mathbf{M}_g$, $\mathbf{M}_i$, $\mathbf{M}_c$	–	–
	Multi-scale Convolution, Feature Fusion	(64, 128), (40, 128), (24, 128)	(64, 128), (40, 128), (24, 128)
	FC ($\mathbf{M}_g$, Concatenation($\mathbf{M}_i$, $\mathbf{M}_c'$))	(64, 128), (40, 128), (24, 128)	(128, 24)
	Channel Attention	(24, 128)	(24, 128)
	Weight Reallocation, Shortcut	(128, 24)	(128, 24)
MoE Downsample Module	MoE Convolution (3,48,2)	(24, 128)	(48, 64)
	LN	(64, 48)	(64, 48)
Output Module	FC (144,576)	(144)	(576)
	GELU	(576)	(576)
	LN	(576)	(576)
	FC (576,1152)	(576)	(1152)
	FC (1152,576)	(1152)	(576)
	FC (576,2)	(576)	(2)

MoE Conv: (experts number, kernel number, stride) **FC:** (input dimension, output dimension).

normalized data is then passed through a fully connected FC layer with a fixed expansion ratio E_r, utilizing the feature map to increase the channel hidden feature dimensions H_d. The resulting hidden feature dimension is calculated as follows:

$$H_d = E_r \times C_{out}. \tag{3}$$

To ensure sufficient channel dimensions for multi-scale feature map segmentation, the final output dimension generated by the FC layer is $2 \times H_d$.

$$\mathbf{M} = \mathbf{y}^T \mathbf{W}_m + \mathbf{b}_1 \in \mathbb{R}^{H' \times 2H_d}. \tag{4}$$

Here, the $\mathbf{y}^T$ corresponds to the transpose of $\mathbf{y}$, and the dimensions of $\mathbf{W}_m$ are given by $\in \mathbb{R}^{C_{out} \times 2H_d}$. The output feature map is divided into three parts along the channel dimension: gating features $\mathbf{M}_g$, information features $\mathbf{M}_i$, and convolution features $\mathbf{M}_c$. The partition indices are $(H_d, H_d - C_{\text{num}}, C_{\text{num}})$, the dimensions of each feature part correspond to the gating feature map $\mathbf{M}_g \in \mathbb{R}^{H' \times H_d}$, information feature map

$\mathbf{M}_i \in \mathbb{R}^{H' \times (H_d - C_{\text{num}})}$, and convolution feature map $M_c \in \mathbb{R}^{H' \times C_{\text{num}}}$. This ensures that the total dimension sums up correctly to 2 H_d. The C_{num} corresponds to the number of output channels in the multi-scale convolution processed in the $\mathbf{M}_c$.

Subsequently, the $\mathbf{M}_c$ feature map is input into the multi-scale convolution architecture, where three distinct scales of convolution operations are applied to adaptively adjust the scale of feature extraction. This architecture is designed to maintain the effectiveness of feature extraction while minimizing the computational complexity of the algorithm as much as possible. Multi-scale convolution operations with kernel sizes of 3, 5, and 7 are applied to capture features at different temporal resolutions, enhancing the model's ability to detect both local details and broader contextual information. The computation process is as follows:

$$\begin{cases} \mathbf{M}_{c3} = \mathbf{W}_3 * \mathbf{M}_c, \\ \mathbf{M}_{c5} = \mathbf{W}_5 * \mathbf{M}_c, \\ \mathbf{M}_{c7} = \mathbf{W}_7 * \mathbf{M}_c. \end{cases} \tag{5}$$

Each operation is performed with the associated convolution kernel $\mathbf{W}_k$, in which k indicates the size of the convolutional kernel employed. The resulting outputs from the convolutions, denoted as $\mathbf{M}_{c3}$, $\mathbf{M}_{c5}$, and $\mathbf{M}_{c7}$, all share the same output dimensions of $\mathbf{M}_c$. The results of the multi-scale convolutions are combined through element-wise averaging, as expressed by the following equation:

$$\mathbf{M}'_c = \frac{\mathbf{M}_{c3} + \mathbf{M}_{c5} + \mathbf{M}_{c7}}{3} \tag{6}$$

This operation corresponds to the multi-scale convolution and feature fusion calculations in the MsA gated CNN module as detailed in Table 1. This approach effectively captures features across different receptive fields, enhancing model sensitivity to features at various spatial scales.

After the feature maps computed through the multi-scale convolution paths are normalized using BN, they are concatenated along the channel dimension with the information feature map $\mathbf{M}_i$. The concatenated feature map is then subjected to element-wise multiplication with the gating features $\mathbf{M}_g$. Subsequently, an FC layer is employed for channel-wise dimensionality reduction to C_r. The calculation process of $\mathbf{M}_u$ is as follows:

$$\mathbf{M}_u = (\text{GELU}(\mathbf{M}_g) \odot \text{Cat}(\mathbf{M}'_c, \mathbf{M}_i))\mathbf{W}_{cr} + b_{cr}. \tag{7}$$

The GELU activation function is used to introduce non-linear information [39]. The $\mathbf{W}_{cr}$ and b_{cr} represent the weights and biases of the FC layer responsible for dimensionality reduction. Given that $(\mathbf{M}_g \odot \text{Cat}(\mathbf{M}'_c, \mathbf{M}_i)) \in \mathbb{R}^{H' \times H_d}$, and $\mathbf{W}_{cr} \in \mathbb{R}^{H_d \times C_r}$, the matrix multiplication yields $\mathbf{M}_u \in \mathbb{R}^{H' \times C_r}$.

After dimensionality reduction, the feature maps undergo a weight reassignment process based on the channel attention mechanism of the squeeze and excitation (SE) block. The input $\mathbf{M}_u$ is subjected to global average pooling to obtain the global information for each channel, denoted as $\mathbf{C}_a \in \mathbb{R}^{C_r \times 1}$. The compression process for the feature map of a single pathway is given by:

$$\mathbf{C}_a = \frac{1}{H'} \sum_{i=1}^{H'} \mathbf{M}_{u,i}. \tag{8}$$

The global channel features $\mathbf{C}_a$ are passed through an FC layer followed by a ReLU activation function to generate the squeeze feature matrix $\mathbf{z}_{cs}$. The reduction ratio of SE block is set to 4. The computation processing is performed as follows:

$$\mathbf{z}_{cs} = \text{ReLU}(\mathbf{C}_a \mathbf{W}_{as} + \mathbf{b}_{cs}), \tag{9}$$

$$\mathbf{s}_{ce} = \sigma(\mathbf{z}_{cs} \mathbf{W}_{ae} + \mathbf{b}_{ce}). \tag{10}$$

The function σ corresponds to the Sigmoid activation function. The channel attention weight matrix $\mathbf{s}_{ce}$ is applied to reweight the channels of $\mathbf{M}_u$. The feature map $\mathbf{M}_u$, after being reweighted by the channel attention mechanism, is element-wise multiplied with the learnable parameters p_a as a gating mechanism. Subsequently, it is summed with the shortcut of the initial input of MsA gated CNN module $\mathbf{y}$. The overall computation process is as follows:

$$\mathbf{O}_M = \mathbf{M}_u \odot \mathbf{s}_{ce} \odot p_a + \mathbf{y}. \tag{11}$$

The $\mathbf{s}_{ce} \in \mathbb{R}^{C_r \times 1}$ is broadcasted along the spatial dimension to match $\mathbf{M}_u \in \mathbb{R}^{H' \times C_r}$ for element-wise multiplication. Similarly, p_a is a learnable scalar that is broadcasted to match the dimensions of $\mathbf{M}_u$.

2.4. MoE downsample module

The computation of the MoE downsample module is achieved by configuring the stride in the MoE convolution. The MoE downsample module is applied to intermediate feature maps during processing and computation. Its primary function is to decrease the spatial resolution further while increasing the number of feature channels, thereby enabling the progressive extraction of more abstract features. The computational process of this model follows a sequence of operations. First, the input feature map undergoes a MoE convolution operation. After the multi-scale expert system extracts features, LN is applied to produce the normalization downsampled output. The resulting computation can be expressed as:

$$\text{LayerNorm}(\text{Conv}_D(I_m)) \tag{12}$$

where I_m represents the intermediate feature map inputs, and Conv_D refers to the MoE convolution. The detailed computational process of the MoE convolution has already been described in the above section, and the LayerNorm corresponds to LN applied to the output. By incorporating a multi-scale expert mechanism, this structure allows the model to adaptively select different convolution kernels based on the input features, enhancing its generalization capability across various scenarios or tasks. It manages computational cost by progressively reducing the spatial dimensions of the input while increasing the number of feature channels to capture more complex and higher-level semantic features.

2.5. Output module

The output layer of MoEF-Net is composed of several FC layers, LN, and activation functions. This module receives feature maps from the MsA Gated CNN in stage 4, defined as the final high-dimensional feature H_f. The first FC layer expands the dimensionality to $\text{fc}_r \times \text{dim}$, where fc_r is a fixed expansion factor, and dim corresponds to the output dimension length of the final MsA gated CNN module. The transformation is described as:

$$H_1 = \text{FC}_1(H_f) \in \mathbb{R}^{\text{fc}_r \times \text{dim}}, \tag{13}$$

where H_1 is the feature map after the first FC transformation. Following this, a GELU activation function and LN are applied:

$$H_2 = \text{LN}(\text{GELU}(H_1)), \tag{14}$$

where H_2 represents the feature map after applying the activation and normalization.

Subsequently, the dimensionality is expanded from $\text{fc}_r \times \text{dim}$ to twice its size to capture a more complex representation of the features. This expansion is performed via a second fully connected layer:

$$H_3 = \text{FC}_2(H_2) \in \mathbb{R}^{2 \times \text{fc}_r \times \text{dim}}. \tag{15}$$

Utilizing the FC bottleneck layer structure, the transformation is then reverted to the original $\text{fc}_r \times \text{dim}$ dimensions:

$$H_4 = \text{FC}_3(H_3) \in \mathbb{R}^{\text{fc}_r \times \text{dim}}. \tag{16}$$

Finally, the transformed feature map H_4 is projected into a two-dimensional output representing the systolic blood pressure (SBP) and diastolic blood pressure (DBP) values:

$$\hat{y} = \mathrm{FC}_4(H_4) \in \mathbb{R}^2, \tag{17}$$

where $\hat{y} = [\hat{\mathrm{SBP}}, \hat{\mathrm{DBP}}]$ is the predicted SBP and DBP values, respectively. Specific model configuration parameters are detailed in the output module section of Table 1.

2.6. The details of training

To handle uneven residual magnitudes and emphasize clinically meaningful discrepancies, we trained with a Differential Weighted Error (DWE) objective. The loss combines an L1 penalty for small residuals and an L2 penalty for larger residuals, switching at a fixed threshold. Formally,

$$\mathcal{L}(\hat{y}_m, y_m) = \frac{1}{N}\sum_{m=1}^{N}\left[\alpha\,\mathbf{I}_{\{|\delta_m|>1\}}\,(\delta_m)^2 + \beta\,\mathbf{I}_{\{|\delta_m|\le 1\}}\,|\delta_m|\right], \tag{18}$$

where N denotes the number of samples, $\delta_m = \hat{y}_m - y_m$ is the prediction error for sample m, and $\mathbf{I}_{\{\cdot\}}$ is the indicator function:

$$\mathbf{I}_{\{|\delta_m|>1\}} = \begin{cases} 1, & |\delta_m| > 1 \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad \mathbf{I}_{\{|\delta_m|\le 1\}} = \begin{cases} 1, & |\delta_m| \le 1 \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

Thus, residuals with $|\delta_m| \le 1$ mmHg incur a linear L1 penalty weighted by β, while residuals with $|\delta_m| > 1$ mmHg incur a quadratic L2 penalty weighted by α. This piecewise design tempers the influence of near-zero noise yet increases curvature and gradients for larger errors, encouraging their correction without overfitting small deviations. We adopt $\alpha = 3.0$ and $\beta = 1.0$, which offered the best overall balance of accuracy and agreement among nearby weightings and are used throughout. Due to considerable interindividual hemodynamic variability, identical blood pressure levels may reflect distinct physiological signal features. To address this, a subset of data was extracted from each sample for model training [33,40]. With a limited sample size, we adopt a 6:4 split for training and testing. Recordings cover resting, Valsalva, apnea and tilt-table conditions, and both splits include these states to encourage cross-state generalization. Optimization used Adam [41]. To improve numerical stability, we applied gradient clipping at 1.0 and Xavier initialization for weights [42]. The learning rate was 1×10^{-4} with a batch size of 32. Training was performed on an NVIDIA RTX 3090 GPU.

3. Experiments and results

3.1. Dataset

A 24 GHz synchronized physiological signal database provided by the Institute of High-Frequency Technology at the Hamburg University of Technology was utilized in this study. The data were collected according to specified protocols by physicians at Erlangen university hospital [43]. The database encompasses various physiological states, including rest, Valsalva maneuver, apnea, and tilt table testing. During tilt table tests, subjects were tilted upwards or downwards to induce changes in cardiopulmonary status. The proportions and durations of these physiological states are presented in Table 2. The diversity of these states provides a basis for studying hemodynamic information across different physiological conditions [44]. Furthermore, the instability of radar echoes generated during these actions poses significant challenges to the reliability of the proposed model across various signal scenarios. The dataset consisted of physiological recordings from 30 healthy subjects, including 14 men and 16 women, with a mean age of 30.7 years, a mean height of 175.7 cm and a mean weight of 72.2 kg. Within the clinical ranges specified in this study, systolic 90–130 mmHg and diastolic 60–90 mmHg, the distribution of measured blood pressures is predominantly normotensive. For systolic pressure, approximately 91.5% of measurements lie within the normal range and 8.5% exceed the upper threshold; systolic hypotension is not observed in this cohort. For diastolic pressure, approximately 85.8% lie within the normal range, 12.0% exceed the upper threshold, and 2.2% fall below the lower threshold.

Table 2
Comprehensive analysis of duration and proportion of various physiological states within the dataset.

Physiological state	Resting	Apnea	Valsalva	Tilt up	Tilt down
Time(min)	86.54	15.93	92.56	52.82	57.86
Proportion(%)	28.31	5.21	30.28	17.28	18.93

Measurements near the upper and lower bounds of the stated normal ranges provide representation of pre-hypertensive and pre-hypotensive states alongside overt hypertension and diastolic hypotension, and this distribution of labels was further used to evaluate the performance of the proposed model in assessing abnormal high and low blood pressure. This composition reflects a healthy volunteer sample acquired across rest, Valsalva, apnea, and tilt-table conditions, and offers coverage around the clinical thresholds. During data collection, radar echo signals were sampled at 2000 Hz, while blood pressure was sampled at 100 Hz. An upsampling interpolation method was employed to expand the blood pressure measurements to the same sampling rate to align the data.

The dataset spans rest, Valsalva, apnea and tilt table. This breadth is used in both training and test partitions to expose the model to state transitions and echo instability during learning and to evaluate behavior on subjects under the same range of conditions.

3.2. Data preprocess

Data preprocessing methods were based on filtering and segmentation techniques proposed in a previous study, which demonstrated effectiveness [26]. Kalman filtering was applied to the original in-phase and quadrature radar channel paths for echo demodulation. The CPMV signal was processed using a 12th-order IIR bandpass filter with a frequency band of 0.8 Hz to 5 Hz, following the filtering range of the PPG signal. Following the second-derivative PPG (SPPG) method, the SCPMV signal was obtained through second-order differencing of the CPMV signal. To isolate the heartbeat component, the raw radar echoes were first band-pass filtered using a fourth-order IIR filter with a passband of 0.9–3 Hz, referred to as the RHB filter, to extract the heartbeat frequency. Signal segmentation was performed every 2.5 s, corresponding to 5000 data points. For network input, each window of CPMV, SCPMV, and RHB was uniformly resampled to 512 points, which yields an effective sampling frequency $f_s = \frac{512}{2.5} \approx 204.8$ Hz and a Nyquist frequency $f_N = \frac{f_s}{2} \approx 102.4$ Hz. By the Nyquist criterion, f_N is sufficient to contain the relevant cardiopulmonary components of interest, so the resampled rate accommodates these signals without aliasing and preserves morphology and timing across channels [45]. The overall signal quality was assessed using a CPMV signal validation algorithm based on multi-lead ECG comparisons. Specifically, two reference heart rates were calculated from ECG signals obtained from two different leads by computing peak intervals. Subsequently, the difference between these two heart rates was compared to assess the reliability of the heart rate calculations from the ECG data. The overall process of data correction using ECG is depicted in the heart rate calculation section in Fig. 2. The relative error of the heart rate, Error_{hr}, is defined as:

$$\mathrm{Error}_{hr} = \frac{|\mathrm{HR}_{\mathrm{radar}} - \mathrm{HR}_{\mathrm{ref}}|}{\mathrm{HR}_{\mathrm{ref}}}. \tag{20}$$

The $\mathrm{HR}_{\mathrm{radar}}$ and $\mathrm{HR}_{\mathrm{ref}}$ denote the heart rate values derived from the RHB signal in the radar data and calculated from the ECG leads,

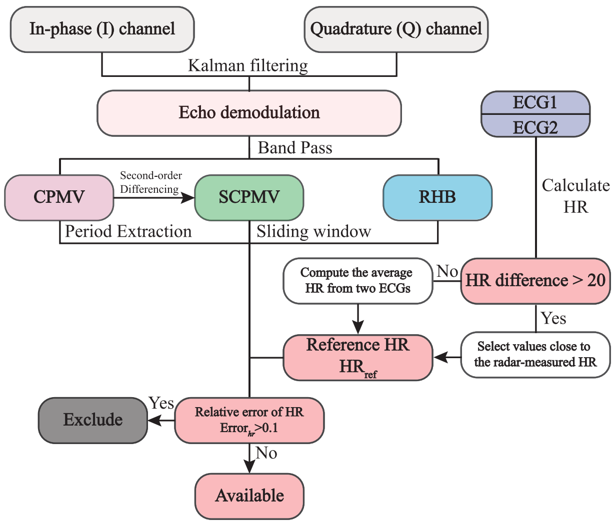

Fig. 2. Data preprocessing methodology from raw radar in-phase and quadrature channels, encompassing signal filtering, heart rate reliability assessment, and signal quality error analysis.

respectively. Data are included in the MoEF-Net dataset when $\mathrm{Error}_{hr} < 0.1$.

Blood pressure readings excessively low or high were considered measurement errors for the ground truth, as all subjects were declared healthy. According to the American Heart Association (AHA) and the American Society of Hypertension (ASH), the normal blood pressure range for healthy adults is defined as a systolic pressure between 90 and 130mmHg and a diastolic pressure between 60 and 90mmHg [46–48]. To account for prehypertension and hypotension, the acceptable blood pressure range for subjects in the experiments was defined as 85 to 150 mmHg for SBP and 55 to 100 mmHg for DBP. A schematic of the complete data preprocessing procedure is shown in Fig. 2.

3.3. Comparison of model performance metrics

Model performance was evaluated using statistical metrics such as Mean Error (ME), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Relative Error (MRE), and Pearson Correlation Coefficient (R). These metrics are calculated as follows:

$$
\begin{aligned}
\mathrm{ME} &= \frac{1}{M}\sum_{m=1}^{M}(y_m - \hat{y}_m)\\
\mathrm{MAE} &= \frac{1}{M}\sum_{m=1}^{M}|y_m - \hat{y}_m|\\
\mathrm{RMSE} &= \sqrt{\frac{1}{M}\sum_{m=1}^{M}(y_m - \hat{y}_m)^2}\\
\mathrm{MRE} &= \frac{1}{M}\sum_{m=1}^{M}\left|\frac{y_m - \hat{y}_m}{y_m}\right|
\end{aligned}
\tag{21}
$$

where M is the total number of samples, y_m is the actual value, and $\hat{y}_m$ is the predicted value.

As shown in Table 3, the performance metrics of various non-contact blood pressure estimation methods were systematically evaluated. Current methodologies primarily rely on machine learning approaches and end-to-end deep learning frameworks specifically designed to extract morphological features associated with hemodynamic characteristics. However, statistical performance comparisons provide only a general reference due to variations in datasets, acquisition methods, and equipment, such as the sensor types and counts, number of test subjects, measurement locations, and radar carrier frequencies. To assess how model performance varies with different input lengths L, performance results were compared across varying input signal lengths. Under conditions of input length $L = 8$, the proposed model achieved optimal performance. Across heterogeneous physiological activities, MoEF-Net consistently outperformed competing methods on primary evaluation metrics. To assess performance across clinical blood-pressure states, we stratified the cohort and report mean absolute error. For systolic blood pressure, MAE was 4.979 mmHg in the hypertensive range and 4.277 mmHg in the normotensive range. No systolic hypotension cases were available. For diastolic blood pressure, MAE was 4.934 mmHg in the hypertensive range, 3.382 mmHg in the normotensive range, and 4.158 mmHg in the hypotensive range. Error magnitude increased toward physiologic extremes, with the largest degradation in systolic hypertension.

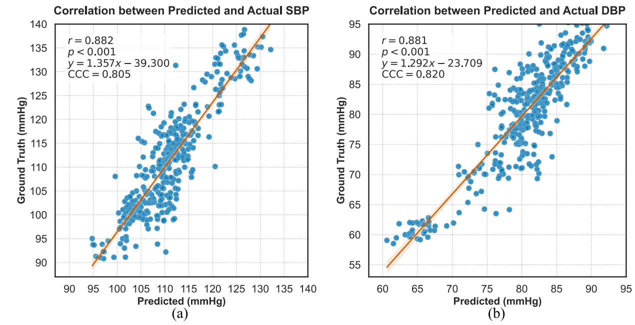

Fig. 3. Correlation between estimated and measured blood pressure values. Scatter plots displaying predicted versus actual SBP (a) and DBP (b). Each subplot presents the Pearson correlation coefficient, p-value, and corresponding linear regression function in the upper left corner.

3.4. Performance evaluation of the international standardized blood pressure assessment

The internationally recognized standards for blood pressure evaluation are those of the Association for the Advancement of Medical Instrumentation (AAMI) and the British Hypertension Society (BHS). Validation under the AAMI protocol requires a minimum of 85 subjects and mandates that the mean error and standard deviation remain within 5 mmHg and 8 mmHg, respectively. As the present cohort comprises 30 healthy volunteers, the study does not qualify for formal AAMI validation, and results are therefore reported solely within the framework of the British Hypertension Society protocol. We report performance under the BHS framework, which evaluates cumulative error distributions. With input length $L = 8$, the method attains Grade A. Detailed cumulative frequencies are provided in Table 4.

3.5. Visualization analysis of statistical performance metrics

Fig. 3 shows scatter plots of model estimates versus reference blood pressure, with panel (a) for SBP and panel (b) for DBP. An ordinary least-squares line overlays each plot to indicate the linear trend. The insets report the Pearson correlation coefficient r and its p-value, together with the fitted relations: SBP $y = 1.357x - 39.300$, DBP $y = 1.292x - 23.709$. These results indicate a strong linear association between predictions and reference values. To quantify agreement, we also report Lin's concordance correlation coefficient, which captures both precision and accuracy by penalizing deviations from the identity line $y = x$. CCC is high for both metrics: SBP 0.805 and DBP 0.820. The slopes exceed unity and the intercepts are negative, consistent with modest scale and offset differences relative to the identity line. The 95% limits of agreement are -11.715 to 11.003 mmHg for SBP and -8.649 to 9.022 mmHg for DBP.

Table 3
Statistical metrics-based performance comparison of models.

Year	Method	Position	Subject	SBP (mmHg)						DBP (mmHg)					
				ME	RMSE	SD	MAE	MRE (%)	R	ME	RMSE	SD	MAE	MRE (%)	R
2019	Linear Regression [49]	wrist & neck	6	−5.3	–	5.1	–	5.10	−0.79	−1.1	–	7.09	–	8.40	−0.82
2020	Reflective PTT [28]	wrist	1	0.55	–	5.45	–	–	–	−2.26	–	3.93	–	–	–
2021	Bagging SVM [50]	chest	8	–	–	–	–	8.30	–	–	–	–	–	8.04	–
2022	Multilayer Perceptron [51]	wrist & chest	27	–	–	12.13	9.51	–	0.78	–	–	8.12	6.69	–	0.77
2022	Linear Regression [52]	chest	4	0.07	6.05	–	–	–	–	−0.07	4.8	–	–	–	–
2022	Random Forest [30]	chest	3	2.52	–	6.73	7.2	–	–	0.22	–	3.85	6.3	–	–
2024	RSD-Net [26]	chest	30	−0.32	6.14	6.14	4.86	4.40	0.84	−0.20	5.50	5.50	4.42	5.69	0.80
2025	R2S-BP [23]	neck & chest	36	−0.16	–	–	5.59	–	0.82	−0.32	–	–	4.10	–	0.82
2025	**MoEF-Net**	**chest**	**30**	**−0.36**	**5.80**	**5.80**	**4.64**	**4.16**	**0.88**	**0.19**	**4.51**	**4.51**	**3.58**	**4.62**	**0.88**

ME:Mean Error; RMSE:Root Mean Square Error
SD:Standard Deviation; MAE:Mean Absolute Error
MRE:Mean Relative Error; R:Correlation Coefficient.

Table 4
Performance of MoEF-Net under BHS standards with varying signal input lengths.

Cumulative frequency of error (%)	<5 mmHg	<10 mmHg	<15 mmHg
Grade A	60	85	95
Grade B	50	75	90
Grade C	40	65	85
SBP prediction (L = 4)	55.1	86.9	97.0
DBP prediction (L = 4)	68.2	93.9	98.9
SBP prediction (L = 6)	58.4	89.3	98.2
DBP prediction (L = 6)	71.9	95.9	100.0
SBP prediction (L = 8)	**62.6**	**91.0**	**98.9**
DBP prediction (L = 8)	**73.8**	**97.0**	**100.0**

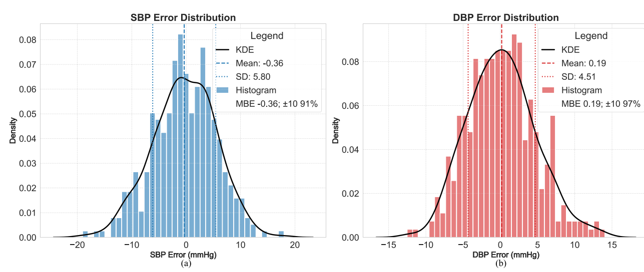

Fig. 4. Prediction errors histogram distribution and kernel density estimate of SBP (a) and DBP (b). Legends corresponding to kernel density estimates, ME, SD, and histograms are located in the upper right corner of each subplot.

Fig. 4 summarizes the error distributions for SBP (a) and DBP (b) using histograms with kernel density estimates. The histograms show the frequency of prediction errors and the KDE curves describe the underlying density. Vertical reference lines mark the sample mean and the ±1 standard deviation, which indicate central tendency and dispersion. The legend reports the mean bias error (MBE) and the proportion of samples with absolute error within ±10 mmHg, providing a compact view of systematic offset and clinical accuracy. Taken together, these descriptors show errors are centered near zero with limited spread and a high fraction of samples within acceptable bounds.

Fig. 5 evaluates the error distributions for SBP (a) and DBP (b) using quantile–quantile (QQ) plots. Each panel compares empirical errors quantiles with the corresponding normal quantiles and overlays a linear fit. Close alignment of the points with this line indicates approximate normality across the central range, while systematic curvature at the ends signals tail deviations. The legend reports the correlation coefficient of the fit line to summarize overall linear conformity. It also reports skewness and excess kurtosis to quantify symmetry and tail weight relative to a normal reference. Low absolute skewness indicates balanced left and right tails, and excess kurtosis near zero indicates tails of normal thickness. Taken together, the visual alignment and the reported statistics provide a compact and rigorous check of distributional adequacy and support the use of parametric summaries in the main analysis.

4. Discussion

The discussion section primarily focuses on ablation experiments conducted on the various modules of MoEF-Net to validate the effectiveness of each structural component. Additionally, the suitability of the corresponding lightweight version for embedded platform deployment is also addressed.

4.1. Ablation experiments

Ablation experiments were conducted to evaluate the various modules of MoEF-Net. The proposed model consists of the MoE stem module and the MoE downsample module, both incorporating a mixture of experts model, and the MsA gated CNN module equipped with channel attention and gating units. Due to their structure of multiple expert components and gating mechanisms, the MoE modules enable adaptive extraction and fusion of input signal features. This allows the model to perform distinct roles within various components by blending feature maps of different depths.

Consequently, ablation experiments were performed by removing the expert mechanisms from the MoE stem module and the MoE downsample module to assess the performance enhancements provided by the MoE modules. Furthermore, in the MsA gated CNN module, channel weight redistribution for $\mathbf{M}_g$, $\mathbf{M}_c$, and $\mathbf{M}_i$ is achieved through attention mechanisms and gating units, enabling adaptive gating of feature information in the final output feature map. The effectiveness of embedding attention and gating mechanisms within the MsA gated CNN module was evaluated by removing these components. The results of the ablation experiments are presented as a heatmap in Fig. 6. Situations S_1, S_2, and S_3 correspond to the removal of the mixture of expert models from the MoE stem module, the MoE downsample module, and the elimination of the channel attention and trainable parameter gating mechanisms in the MsA gated CNN module. As shown in Fig. 6, the MAE and RMSE metric for DBP is most affected in the ablation experiments, resulting in a reduction in performance. The results demonstrate the effectiveness of the various embedded modules within MoEF-Net.

4.2. Module lightweighting

To enhance compatibility with offline hardware and edge computing platforms, a structural rearrangement strategy was applied to the feature extraction stages, leveraging the modularity of the architecture. This reconfiguration involved adjustments to both the depth and channel dimensions across the network, significantly reducing model size and computational demand. The initial stage retained its single-layer gating design to preserve essential feature extraction capability.

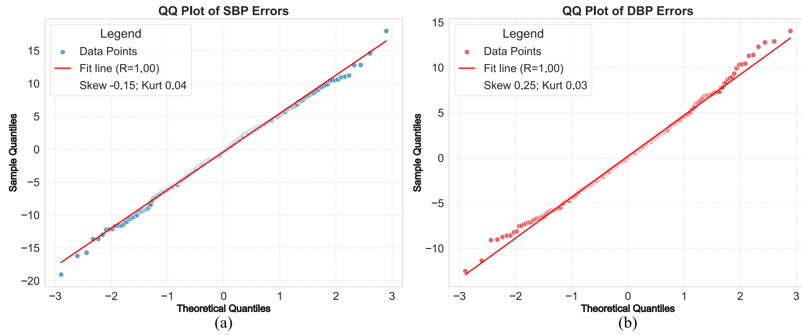

Fig. 5. Quantile–Quantile (QQ) plots of SBP (a) and DBP (b) prediction errors. Legends corresponding to data points and standard correlation coefficient R regression lines are located in the upper left corner of each subplot.

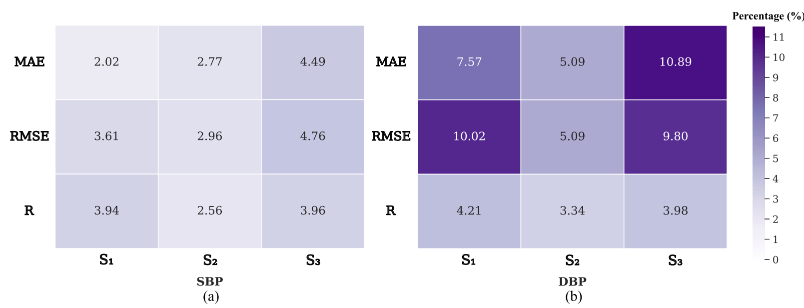

Fig. 6. The ablation results of the MoEF-Net model were evaluated using SBP (a) and DBP (b). Situations S_1, S_2, and S_3 correspond to the percentage decrease in performance following the removal of the mixture expert models from the MoE stem module and MoE downsample module, respectively, and to the extent of performance degradation resulting from the elimination of the attention and gating mechanisms in the MsA gated CNN module.

Table 5
Parameters (M), GFLOPs (M), Computation time (ms) and corresponding performance of MoEF-Net and its lightweight versions.

Model	Params (M)	GFLOPs (M)	Computation time (ms)	SBP (mmHg)						DBP (mmHg)					
				ME	RMSE	SD	MAE	MRE (%)	R	ME	RMSE	SD	MAE	MRE (%)	R
MoEF-Net	2.99	0.0459	30.38 ± 0.92	−0.36	5.80	5.80	4.64	4.16	0.88	0.19	4.51	4.51	3.58	4.62	0.88
MoEF-Net$_L$	1.69 (**−43.48%**)	0.0073 (**−84.10%**)	12.86 ± 0.38 (**−57.67%**)	0.48	6.00	6.00	4.74	4.26	0.86	0.88	4.67	4.67	3.71	4.86	0.86

In subsequent stages, the depths were reconfigured: the second and third stages were adjusted to a depth of 6 each, compared to the original 3 and 9, while the fourth stage maintained a depth of 1. This redistribution allows the model to balance learning of shallow and deep features, minimizing reliance on excessive depth and associated computational costs. Additionally, the number of channels in the first three stages was reduced uniformly to 12, down from the original 24, 48, and 96. This compression strategy emphasizes feature reuse across layers, maintaining informational sufficiency with fewer channels while preserving representational capacity through nonlinear transformations. The final stage retained a width of 144 channels to ensure high-dimensional feature availability for robust prediction.

It is noteworthy that while the original MoEF-Net sets a benchmark for estimation accuracy, the subsequent analysis of its lightweight variant reveals that the minimal performance gain comes at the cost of an increase in computational complexity. Therefore, the overall performance evaluation should prioritize the practical utility achieved by the lightweight model, which delivers comparable accuracy with far greater efficiency. The evaluation encompassed parameter count, computational complexity measured in giga floating point operations (GFLOPs), and forward pass time, with deployment verified on physical hardware. The lightweight variant MoEF-Net$_L$ achieves a reduction in computational demand, including parameter count, GFLOPs, and latency, at the cost of only a marginal loss in performance. As evidenced in Table 5, it maintains practical inference speeds of tens of milliseconds while requiring significantly fewer resources. This favorable trade-off confirms that MoEF-Net$_L$ is well-suited for real-time blood pressure estimation and embedded deployment, leveraging its modular design for high efficiency.

5. Conclusion

This study proposed a millimeter wave radar-based homogeneous multi-modal method for input and concatenating physiological signals, along with the corresponding blood pressure estimation algorithm MoEF-Net model. The CPMV, SCPMV, and RHB signals containing hemodynamic information were utilized as the physiological signal input dataset graph information through radar platforms. Feature maps of biomedical information from these three pathways were concatenated along the channel dimension to form the final input data format. This data processing method converted the multi-dimensional single input signals into three-channel feature maps, enabling the model to utilize processing methods from the time domain waveform detail. The MoEF-Net model, inspired by the metaformer architecture, was designed with phased modules. The structure includes input layer configurations and down-sampling layers that integrate adaptive MoE mechanisms. Model parameters were adaptively adjusted through the gated unit mechanism for the fusion and extraction of feature maps at varying depths. The model scored an A grade under the international BHS standard for blood pressure estimation.

Furthermore, the developed lightweight variant MoEF-Net$_L$ exemplifies the practical utility of our modular framework. It achieves an optimal trade-off by reducing computational costs by over 84% while limiting the performance loss to less than 0.13 mmHg in MAE.

Given that this negligible accuracy sacrifice enables vastly superior deployment efficiency, we recommend MoEF-Net$_L$ as the model of choice for most real-world clinical applications. The original MoEF-Net serves as an accuracy benchmark, but the lightweight version provides the advantage for implementation in resource-constrained medical environments. Expert system components mitigate errors arising from physiological signal instability, with the mixture of experts and gating improving robustness to contactless variability. The method shows stable BHS metrics across input lengths and reaches Grade A at $L = 8$, and residual diagnostics are consistent with well behaved errors. Future work will expand external validation to additional devices, sites and populations, and will explore larger model capacity and feature adaptive selection through enhanced MoE units to further improve performance.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ye Qiu: Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Zhenmiao Deng:** Writing – review & editing, Supervision, Project administration, Investigation. **Xiaohong Huang:** Supervision, Project administration, Investigation. **Shaocan Fan:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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